

Marginal information and acquisition

Rounds 1–2: proof appendix and Week 15 cross-audit

Research working note; priority not established

4 October 2026

Abstract

For a finite binary pool, expected Hamming-error reduction is determined by first moments and uncentered pair moments. This note proves the sharp one-query margin guarantee, the balanced marginal-only minimax value, an exact formula for homogeneous rare-class marginals, and misspecification bounds. The general sharp finite-horizon margin constant remains unresolved. It audits Week 15 and distinguishes coherent conditioning from Laplace refitting. This document and all computations are outside the repository; no empirical acquisition run was repeated.

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1 Scope, status, and priority audit

All laws are conditional on the current data unless stated otherwise. The default model has a fixed pool of $N \geq 2$ binary labels, noiseless revelation of the queried label, equal-weight evaluation on *all* N points including queried points, and ideal conditioning under one joint law. Each realized

world may be deterministic while the unknown label vector is random epistemically. A spin takes values $\{-1, +1\}$; transform 0/1 labels before using spin formulas. Set $\text{sgn}_+(0) = +1$.

Ratios compare expected *risk reductions*, not residual errors. Zero optimal gain gives a vacuous guarantee. Hamming loss is not NSD, ASSD, or boundary Dice. Statements about other targets, noise, weights or model updates explicitly change the default assumptions. **[PROVED]** means proved here, **[COUNTEREXAMPLE]** a refuted generalization with a construction, **[NUMERICALLY CHECKED]** a finite computation, and **[CONJECTURE]** an unresolved claim. Mathematical correctness does not establish publication priority.

1.1 Closest results and verdicts

This is a bounded primary-source audit, not a certificate of absence.

Source	Closest result	Verdict
Mussmann–Liang, ICML 2018 [1]	Theorems 4.1–4.2: asymptotic uncertainty-sampling data efficiency versus passive logistic regression.	Different comparator and regime; not a fixed-pool one-step EER ratio.
Mussmann–Liang, NeurIPS 2018 [2]	Uncertainty sampling as preconditioned SGD on smoothed zero-one loss.	Optimization/stationarity results, not the $1/(N-1)$ constant.
Mussmann et al., 2022 [3]	Section 3.2, PDF equation (6), MEZL: expected zero-one loss via pairwise predictive probabilities.	Direct priority for pairwise EER. Our binary max identity is a specialization.
Liu–Li, 2023 v1 [4]	Equivalent losses and excess-risk guarantees for uncertainty weighting.	Different optimization model; no matching finite-pool minimax theorem located. Do not attribute later revisions to the original version.
Bickford Smith et al., EPIG 2023 [5]	Section 4: predictive mutual information targeted to test inputs, and expected cross-entropy after querying.	Direct priority for target-aware dependence; not zero-one sharp constants.
Hübotter et al., 2024 [6]	Conference Theorems 3.2–3.3: transductive variance/confidence bounds with distinct sampling and target sets.	Structural transfer guarantees for function observations; not binary Hamming ratios without extra assumptions.
Wang–Sun–Grosse, 2021 [7]	Predictive-correlation benchmarking through TAL, metacorrelations, and XLL.	Strong conceptual priority for the failure of marginal quality to imply acquisition quality; also separates selection from prediction models.
Pass–Spektor [8]	Extremal moments of pairwise independent Rademacher sums and exchangeable sign-slice constructions.	Close extremal method, different objective. Ingredients are not new.
Cut-polytope gap inequalities [9]	Integer/parity strengthening of positive semidefiniteness.	The parity step for κ_N is a known gap inequality.
Bickford Smith–Foster–Rainforth 2024 [10]	Section 4.2.1: mismatch between assumed Bayesian updates and actual learning.	Direct conceptual priority for the update-consistency caveat.
Hu–Mussmann, arXiv v3, 2026 [11]	Myopic Bayesian decision theory and partial batch label sampling for EPIG.	Further decision-theoretic prior art; not the optimal adaptive Hamming ratio.

No equivalent exact $1/(N-1)$ or κ_N statement was located in these sources or targeted searches under EER competitive ratios, cut-matrix norm ratios, sign-sum extremality, or oblivious/marginal-only selection. *Priority remains unestablished.* The general claim that marginals miss acquisition-relevant dependence is already established prior art. The rare-marginal results below also need a separate priority audit.

2 Binary identity and sharp one-step margin bound

Theorem 2.1 (Arbitrary binary target and observation). *[PROVED] Let $T, O \in \{-1, +1\}$ have any joint law, $m = \mathbb{E}T$, $c = \mathbb{E}TO$. The minimum expected zero-one error after observing O is*

$$e(T | O) = \frac{1 - \max\{|m|, |c|\}}{2}.$$

For targets T_i , weights $w_i \geq 0$, $\sum_i w_i = 1$, and observations O_j , put $m_i = \mathbb{E}T_i$ and $c_{ij} = \mathbb{E}T_i O_j$. Then

$$R_0 = \frac{1}{2} \sum_i w_i (1 - |m_i|), \quad (1)$$

$$F(j) = \frac{1}{2} \sum_i w_i [1 - \max\{|m_i|, |c_{ij}|\}], \quad (2)$$

$$\Delta(j) = \frac{1}{2} \sum_i w_i (|c_{ij}| - |m_i|)_+. \quad (3)$$

Proof. $\mathbb{E}[T \mathbf{1}_{\{O=s\}}] = (m+sc)/2$. Optimal accuracy minus error therefore equals $\sum_s |\mathbb{E}[T \mathbf{1}_{\{O=s\}}]| = (|m+c| + |m-c|)/2 = \max(|m|, |c|)$. Zero-probability branches cause no problem. Sum the target risks. $\square$

Theorem 2.2 (Sharp margin guarantee). *[PROVED] In the default model put $u_i = \min(p_i, 1 - p_i)$ and choose $a \in \arg \max_i u_i$. Then $\Delta(a) \geq \max_j \Delta(j)/(N-1)$. The infimum is sharp even with a unique margin maximizer. For $N = 2$, margin is optimal.*

Proof. Let $e_i^{(j)}$ be expected posterior error, $v_{ij} = u_i - e_i^{(j)}$ and $G_j = \sum_i v_{ij} = N\Delta(j)$. Then $0 \leq v_{ij} \leq u_i$, $v_{jj} = u_j$, and $u_a \geq u_b$ implies $e_b^{(a)} \leq e_a^{(b)}$ by Theorem 2.1. Thus $u_b + v_{ab} \leq u_a + v_{ba}$ and

$$G_b \leq u_b + v_{ab} + (N-2)u_a \leq G_a + (N-2)u_a \leq (N-1)G_a.$$

For sharpness take an independent fair decoy D and $N-1$ copies of $Z \sim \text{Bernoulli}(1/2 + \eta)$. Margin uniquely queries D . Gains are $1/(2N)$ and $(N-1)(1/2 - \eta)/N$, whose ratio tends to $1/(N-1)$. $\square$

Corollary 2.3 (Independence, entropy, perturbations). *[PROVED] Pairwise independence makes margin one-step Hamming-optimal. Noiseless margin maximizes reduction of joint label entropy for every joint law. If $|\hat{m}_i - m_i| \leq \epsilon_m$ and $|\hat{c}_{ij} - c_{ij}| \leq \epsilon_c$, each score (3) has error at most $(\epsilon_m + \epsilon_c)/2$, and its estimated maximizer has regret at most $\epsilon_m + \epsilon_c$.*

Proof. Under independence $c_{ij} = m_i m_j$ for $i \neq j$, so off-diagonal gains vanish. For entropy, $I(Y; Y_j) = H(Y_j)$, increasing with u_j . For perturbations use the Lipschitz properties of absolute value and positive part, then compare the estimated and true maximizing scores. $\square$

3 Balanced minimax and the cut polytope

Theorem 3.1 (Exact balanced marginal-only minimax). *[PROVED] Suppose all $p_i = 1/2$. A marginal-only selector uses probabilities q_j , without access to the joint law. Put $M = N-1$; let a be the greatest nonnegative integer at most $\sqrt{M}$ with the same parity as M , and $b = a+2$. Set*

$$\tau = (M + ab)/(a + b), \quad \kappa_N = \frac{N + 2\tau}{N(1 + \tau)}.$$

Then uniform selection is minimax and

$$\max_{q \in \Delta_N} \inf_P \frac{\sum_j q_j \Delta_P(j)}{\max_j \Delta_P(j)} = \kappa_N.$$

Equivalently, with $\mathcal{C}_N = \text{conv}\{ss^\top : s \in \{-1, +1\}^N\}$,

$$\min_{C \in \mathcal{C}_N} \frac{\sum_{ij} |C_{ij}|}{N \max_k \sum_i |C_{ik}|} = \kappa_N.$$

Proof. Global sign symmetrization makes means zero without changing second moments, so $\mathcal{C}_N$ is exactly the feasible set. Let a maximal column k have sum $1 + t$ and let $A = \sum_{ij} |C_{ij}|$. Diagonal and root entries imply $A \geq N + 2t$. Let $W = \sum_{i \neq k} \sigma_i S_i S_k$ with $\sigma_i C_{ik} = |C_{ik}|$. Then $\mathbb{E}W = t$ and W lies on the parity- M lattice, so $(W - a)(W - b) \geq 0$. Consequently

$$\mathbb{E}W^2 \geq (a + b)t - ab, \quad A \geq 1 + 2t + \mathbb{E}W^2 \geq 1 + (2 + a + b)t - ab.$$

Divide by $N(1 + t)$: the first bound decreases and the second increases. Their intersection is $t = \tau$, yielding the lower bound.

For attainment take a fair root R and an independent sign vector V of length M . Its sum W takes values a, b with $\mathbb{P}(W = b) = (M - a^2)/(b^2 - a^2)$; conditional on W , use a uniform sign slice. Set root spin R and leaf spins $-RV_i$. Then $\mathbb{E}W = \tau$, $\mathbb{E}W^2 = M$, leaf-pair correlations vanish, and root-leaf correlations equal $-\tau/M$. The root and leaf column sums are $1 + \tau$ and $1 + \tau/M$. Uniform selection attains the bound. For any q place the root at an index with $q_k \leq 1/N$, making its ratio no larger than the uniform ratio. Every world in this construction contains both classes. $\square$

Proposition 3.2 (PSD relaxation and known gap inequality). *[PROVED] Over all PSD unit-diagonal matrices the same minimum is $(1 + \sqrt{N - 1})/N$. It equals κ_N exactly when $N - 1$ is a square. The parity step above is a cut-polytope gap inequality.*

Proof. The Schur complement gives the algebraic analogue $\mathbb{E}W^2 \geq t^2$, hence the ratio is at least $\max\{(N + 2t)/(N(1 + t)), (1 + t)/N\}$. These intersect at $t = \sqrt{M}$. The matrix with leaf block I_M and every root-leaf entry $1/\sqrt{M}$ is PSD and attains the bound. For $N > 2$, binary equality requires $W = \sqrt{M}$ almost surely, possible only for square M ; the extremal slice construction then attains it. The case $N = 2$ has value one directly. For the gap interpretation take integer coefficients $b_k = -(a + 1)$, $b_i = \sigma_i$. The odd integer $b^\top S = S_k(W - a - 1)$ satisfies $(b^\top S)^2 \geq 1$, precisely $(W - a)(W - a - 2) \geq 0$. These inequalities predate this note [9]. $\square$

[NUMERICALLY CHECKED] Round-1 checks of extremal laws for $N = 4, \dots, 10$ agree with this proof. Examples: $\kappa_3 = 5/6$, $\kappa_4 = 7/10$, $\kappa_5 = 3/5$, $\kappa_6 = 5/9$, $\kappa_7 = 1/2$, $\kappa_{10} = 2/5$.

4 General marginals and an exact rare-class formula

Proposition 4.1 (Fully oblivious versus marginal-only selection). *[PROVED] A selector fixed before seeing either marginals or dependence has minimax gain ratio $1/N$ over all binary pool laws. Uniform attains it. This is a different game from the balanced marginal-only game. For any gain guarantee $\Delta_a \geq r\Delta^*$, the corresponding residual-risk statement is $R_a \leq (1 - r)R_0 + rR^*$, not $R_a/R^* \leq 1/r$.*

Proof. Uniform averages nonnegative gains, hence achieves at least their maximum divided by N . For any fixed selector choose its least-probable coordinate as an independent fair bit and make every other label deterministic. Only that query has nonzero value, giving ratio at most $1/N$. If all marginals must be nondegenerate, approximate the deterministic ones by independent bits of vanishing uncertainty. The risk conversion follows by subtracting gains from R_0 . $\square$

Theorem 4.2 (Finite LP characterization). *[PROVED] Fix $p = (p_1, \dots, p_N)$ with some non-degenerate coordinate. Let $\mathcal{P}(p)$ be all world laws with these marginals. Partition this probability simplex by $c_{ij} = \pm|m_i|$, and collect the vertices v of all cells. Set $G_j(v) = N\Delta_v(j)$ and $r_j(v) = G_j(v)/\max_k G_k(v)$. The minimax value and an optimal selector solve*

$$\max_{q, z} z, \quad q_j \geq 0, \quad \sum_j q_j = 1, \quad \sum_j q_j r_j(v) \geq z \quad \text{for every cell vertex } v.$$

An optimizer can be constant within each identical-marginal group. For two levels, selection therefore reduces to one group-mass variable, although enumerating constraints remains exponential.

Proof. Each G_j is linear on each cell. For fixed q , the ratio equals $\min_{k: G_k > 0} (q^\top G)/G_k$. A linear-fractional function with nonnegative denominator attains its infimum at a positive-denominator vertex: express any point as a convex combination of vertices and weight their ratios by their denominators. Zero-denominator vertices add a nonnegative numerator. At every law some $G_j \geq u_j > 0$. Taking all cells proves the LP. The worst-law value is concave in q and invariant under within-group permutations; averaging an optimizer proves the symmetry statement. $\square$

Proposition 4.3 (Uniform is not always minimax). *[PROVED] If all $p_i = p \in (0, 1)$, uniform selection is minimax; the same holds when all u_i are equal, after known coordinate label flips. For $N = 2$, $u_1 \geq u_2 > 0$, the minimax value is one, attained by margin. A selector choosing point 1 with probability q has worst-case ratio $q + (1 - q)u_2/u_1$.*

Proof. Average over all permutations in the homogeneous case; flips preserve risks. For $N = 2$, margin is optimal. Let $e_i^{(j)}$ be expected posterior errors. Then $e_1^{(2)} \geq e_2^{(1)}$, and $G_1 - G_2 = e_1^{(2)} - e_2^{(1)} \leq u_1 - u_2$. The last bound follows directly by subtracting the two max formulas. Since $G_1 \geq u_1$, $G_2/G_1 \geq u_2/u_1$. Independent labels attain equality. $\square$

[COUNTEREXAMPLE] At $(p_1, p_2) = (1/2, 1/100)$, uniform has value 0.51 while margin has value one. Thus heterogeneous imbalance can make uniform suboptimal.

Theorem 4.4 (A nontrivial exact two-level family). *[PROVED] For $N = 3$, marginals (a, a, b) , $0 < 2a \leq b \leq 3a$ and $b \leq 1/3$, put $r = b/a \in [2, 3]$. An optimal marginal-only selector assigns total mass*

$$\theta = \frac{r(3-r)}{6r-3-r^2}$$

equally to the first two points and mass $1 - \theta$ to the third. Its exact minimax value is

$$V(a, a, b) = \frac{2r}{6r-3-r^2}.$$

In particular, for $(a, a, b) = (.1, .1, .25)$, the selector is $(5/46, 5/46, 18/23)$ and its value is $20/23$. Pure margin has value $5/6$ in this example; uniform has value $3/5$.

Proof. All probabilities are at most $1/3$, so the influence of query j on target i is $(2q_{ij} - p_j)_+$, where $q_{ij} = P(Y_i = Y_j = 1)$. Since $b \geq 2a$, $G_3 = b$ always. Put $x = (2q_{12} - a)_+$ and $y_i = (2q_{i3} - a)_+$. Then $G_1 = a + x + y_1$, $G_2 = a + x + y_2$, with $0 \leq x, y_i \leq a$. Symmetry permits a selector with masses $(\theta/2, \theta/2, 1 - \theta)$. Independent labels give ratio $1 - \theta(1 - 1/r)$. Taking $Y_1 = Y_2$ with their common event a subset of $\{Y_3 = 1\}$ gives ratio $\theta + (1 - \theta)r/3$. Balancing these two lines gives the asserted θ and an upper bound on every selector's worst-case ratio.

For the matching lower bound, if $\max(G_1, G_2) \leq b$, the ratio is at least the independent-label line since each $G_i \geq a$. Otherwise suppose $d = G_1 > b \geq 2a$. Both x, y_1 are positive and $q_{23} \geq q_{12} + q_{13} - a$ implies $y_2 \geq x + y_1 - a = d - 2a$. Also $x \geq d - 2a$, so $G_2 \geq 2d - 3a$. The ratio is therefore at least

$$\frac{\theta(3d - 3a)/2 + (1 - \theta)b}{d}.$$

For the specified θ , $0 \leq \theta \leq 2/5$ and $b(1 - \theta) - 3a\theta/2 > 0$. This expression decreases with $d \leq 3a$, and its value at $3a$ is exactly the second line. This proves the minimax formula. For pure margin $b/\max G \geq b/(3a)$, attained by the nested law. For uniform in the stated example, if the maximum is b the ratio is at least $(2a + b)/(3b) = 3/5$. If it is $d > b$, the preceding bound gives $1 + (b - 3a)/(3d) > 3/5$. Independence attains $3/5$. $\square$

Theorem 4.5 (Exact homogeneous rare-class value). *[PROVED] Suppose $p_i = p$, $0 < p \leq 1/3$, and denote the minimax value by $\kappa_N(p)$. For $z \geq 0$ define $\phi(z) = 2\lfloor z \rfloor z - \lfloor z \rfloor(\lfloor z \rfloor + 1)$. For $1 \leq K \leq N - 1$, $0 \leq t \leq K$, put*

$$\mu_1 = (K + t)/2, \quad \mu_0 = \frac{p(K - t)}{2(1 - p)}, \quad L_{K,p}(t) = (2p\phi(\mu_1) + 2(1 - p)\phi(\mu_0) - pK(K - 1))_+.$$

Then

$$\kappa_N(p) = \min \left\{ 1, \min_{\substack{1 \leq K \leq N-1, \ 0 \leq t \leq K \\ L_{K,p}(t) \leq p(K-1)t}} \frac{Np + 2pt + L_{K,p}(t)}{Np(1 + t)} \right\}. \quad (4)$$

Only finitely many endpoints need checking: floor breakpoints and crossings of the piecewise-linear constraints.

Proof. Write $q_{ij} = \mathbb{P}(Y_i = Y_j = 1)$. The influence is $v_{ij} = (2q_{ij} - p)_+$, including $v_{ii} = p$. The negative-correlation branch can contribute only if $q_{ij} < (3p - 1)/2$, impossible here. Choose a maximal-gain root with row sum $p(1 + t)$ and let its K active neighbors satisfy $q_{i0} > p/2$. If $t = 0$ all rows equal p . Otherwise let Z count the ones in the active set. Summing root joint probabilities gives conditional means μ_1, μ_0 . The convex envelope of $k(k - 1)$ on integers is ϕ , so $\mathbb{E}[Z(Z - 1)] \geq p\phi(\mu_1) + (1 - p)\phi(\mu_0)$. The ordered active-leaf influence sum is at least $L_{K,p}(t)$, and the full sum is at least $Np + 2pt + L_{K,p}(t)$. Sum the K inequalities saying each active leaf row is at most the root row: its active-leaf influence sum is at most $p(K - 1)t$. Thus every law yields a feasible pair and the lower bound in (4).

Conversely take a Bernoulli(p) root. Conditional on root s , draw Z on the two neighboring integers around μ_s with that mean, then choose a uniform subset of Z active leaves to label one. Every active leaf has marginal p and root influence pt/K . Exchangeability makes its total ordered leaf influence exactly $L_{K,p}(t)$. Take remaining labels independently Bernoulli(p), independent of the block; their cross-influences vanish. The feasibility constraint is exactly that the root row be maximal. This attains every feasible displayed ratio. Between all specified breakpoints the objective is linear fractional and is minimized at an endpoint (or is constant). $\square$

Proposition 4.6 (Imbalance changes the asymptotic order). *[PROVED] For every fixed $0 < p < 1/2$, $\kappa_N(p) = \Theta(N^{-1})$. Define $t_p = 2p - 1 + \sqrt{2(1 - p)(1 - 2p)} > 0$. Then*

$$\frac{3N - 2}{N^2} \leq \kappa_N(p) \leq \frac{N + 2(N - 1)t_p}{N[1 + (N - 1)t_p]}.$$

For $0 < p \leq 1/3$, $\lim_{N \rightarrow \infty} N\kappa_N(p) = 2 + t_p^{-1}$.

Proof. Homogeneous influences are symmetric, nonnegative, bounded by p . If their maximal row is D , the total is at least $Np + 2(D - p)$ and $D \leq Np$. Divide by ND . For the upper bound use a root of probability p and conditionally iid leaves with probabilities $a = p + \sqrt{(1 - p)(1/2 - p)}$ given root 1 and $b = p(1 - a)/(1 - p)$ given root 0. Leaves have marginal p ; distinct leaf pairs have joint probability $p/2$ and zero influence. Root-leaf influence is $p(2a - 1) = pt_p$.

For the limit, an optimizing pair in (4) has $t = \Omega(N)$, $K \geq t = \Omega(N)$ and $L/p = O(N)$, by the upper bound and numerator Np . The inequality $\phi(z) \geq z^2 - z$ yields

$$L/p \geq [K^2 f_p(t/K) - K]_+, \quad f_p(z) = \frac{z^2 + 2(1 - 2p)z - (1 - 2p)}{2(1 - p)}.$$

This increasing quadratic for $z \geq 0$ has positive root t_p . Thus $t/K \leq t_p + o(1)$, so $t/N \leq t_p + o(1)$. Finally $N\kappa_N(p) \geq (N + 2t)/(1 + t) = 2 + (N - 2)/(1 + t)$ gives the matching lower limit. $\square$

[NUMERICALLY CHECKED] Exact rational endpoint enumeration for $N = 136$, $p = 3/34 = 12/136$ gives $\kappa_{136}(3/34) = 2798/86615 \simeq 0.0323038735$. An attaining law has $K = 135$, $t = 5004/91$, $L = 0$, $\mu_1 = 17289/182$, $\mu_0 = 21843/5642$, and leaf-pair probability $3/68 = p/2$. This fixes each marginal to $12/136$; it does not fix the realized count. Under the uniform law on exactly 12 rare labels all query values coincide, so the ratio is one. The finite LP can incorporate a fixed-count constraint.

5 Different targets and noisy observations

Proposition 5.1 (Exact positivity condition). *[PROVED] For arbitrary targets and binary observations, query a has positive Bayes Hamming value iff some positive-weight target satisfies $|c_{ia}| > |m_i|$. If a set of targets of total weight at least ω has $|c_{ia}| - |m_i| \geq g > 0$, then*

$$\Delta(a) \geq \omega g/2, \quad \Delta(a)/\Delta^* \geq \omega g/(2R_0) \geq \omega g.$$

Proof. Use the nonnegative summands in (3) and $\Delta^* \leq R_0 \leq 1/2$. $\square$

This is a necessary-and-sufficient pointwise condition for positive value, and a quantitative sufficient condition for a ratio. There is no unique weakest named family of structural assumptions; any uniform ratio needs quantitative transfer of this kind or an equivalent restriction.

Counterexample 5.2 (Disjoint targets). **[COUNTEREXAMPLE]** A fair target T and a fair candidate O_1 can be independent, while O_2 is a noisy copy of T with $\mathbb{E}O_2 \neq 0$ and positive information value. For example set $O_2 = +1$ when $T = +1$ and, when $T = -1$, let $O_2 = +1$ with probability $1/2$. Then $\mathbb{E}O_2 = 1/2$, $\mathbb{E}TO_2 = 1/2$. Margin uniquely selects O_1 , with zero value, whereas O_2 has value $1/4$. Hence target–pool separation admits no unconditional positive margin ratio.

Theorem 5.3 (Common binary symmetric observation noise). *[PROVED] In the full-pool Hamming model observe $O_j = S_j \xi_j$, where the one-query channel noise is independent of the world and $\mathbb{E}\xi_j = \eta = 1 - 2\epsilon \in [0, 1]$, common to all candidates. Latent-label margin still satisfies $\Delta(a) \geq \Delta^*/(N - 1)$. For $\eta > 0$ the constant is sharp. Observation-marginal margin has the same ranking as latent margin.*

Proof. Now $v_{ij} = (\eta|c_{ij}| - |m_i|)_+/2$ and $v_{ii} = w_i = (\eta - |m_i|)_+/2 = (u_i - \epsilon)_+$. For every i, j , $0 \leq v_{ij} \leq w_i$, and

$$w_i - v_{ij} = \frac{1}{2}[\eta - \max\{|m_i|, \eta|c_{ij}|\}]_+.$$

If $|m_a| \leq |m_b|$, the right side at (b, a) is no larger than at (a, b) . Repeat the proof of Theorem 2.2 with w_i in place of u_i . The independent fair decoy and slightly biased duplicate cluster approach the same sharp ratio. Finally $|\mathbb{E}O_j| = \eta|m_j|$. $\square$

Counterexample 5.4 (Informative but heterogeneous noise). **[COUNTEREXAMPLE]** Take independent labels with probabilities $(0.10, 0.09)$ and BSC error rates $(0.20, 0.01)$. Both latent and observation margin choose point 1. Its Bayes value is zero, because its uncertainty 0.10 is below its channel error 0.20. Query 2 has full-pool gain $(0.09 - 0.01)/2 = 0.04$. Every channel is informative; that alone does not guarantee a positive ratio.

Corollary 5.5 (Weighted overlapping targets). *[PROVED] If the targets are the pool labels, possibly with known sign flips, and all target weights lie in $[w_{\min}, w_{\max}]$ with $w_{\min} > 0$, then noiseless or common-BSC margin has ratio at least $w_{\min}/[w_{\max}(N - 1)]$.*

Proof. Sandwich each weighted sum of nonnegative influences between $w_{\min}$ and $w_{\max}$ times its unweighted sum. $\square$

6 Sequential budgets: proved bounds and unresolved sharp constant

Theorem 6.1 (Finite-horizon bounds). *[PROVED] In the default noiseless model, sequential margin with exact updating and B distinct queries satisfies*

$$G_{\text{margin}}(B) \geq (B/N)R_0 \geq (B/N)G_{\text{opt}}(B), \quad 1 \leq B \leq N,$$

where the comparator is the optimal adaptive B -step policy. For $B = 1$ the sharper ratio is $1/(N - 1)$; for $B = N$ it is one. For $N \geq B + 2$ the worst-case ratio is at most $B/(N - 1)$. For fixed B and $N \rightarrow \infty$ these bounds establish the sharp leading order B/N .

Proof. After t queries, posterior Bayes risk is the sum of the $N - t$ remaining uncertainties divided by N . Its largest uncertainty is at least $NR_t/(N - t)$; observing that point reduces risk by at least its own uncertainty divided by N . Thus $\mathbb{E}[R_{t+1} | H_t] \leq (1 - 1/(N - t))R_t$. Iterating gives $\mathbb{E}R_B \leq (N - B)R_0/N$.

For the upper construction take B independent fair decoys and $M = N - B$ copies of an independent Bernoulli($1/2 + \eta$) bit. Margin spends all B queries on decoys. For small η , an optimal policy queries the cluster and $B - 1$ decoys; further cluster queries are useless. The ratio is $B/[M(1 - 2\eta) + B - 1] \rightarrow B/(N - 1)$. $\square$

Counterexample 6.2 (The tempting $B/(N - 1)$ lower bound is false). **[COUNTEREXAMPLE]** For $N = 3$, $B = 2$, let

$$P(010) = 1/6, \quad P(100) = 1/3 - \varepsilon, \quad P(101) = 1/3 + \varepsilon, \quad P(110) = 1/6, \quad 0 < \varepsilon < 1/6.$$

Uncertainties are $(1/6, 1/3, 1/3 + \varepsilon)$, so margin first queries coordinate 3. If it is zero, margin queries coordinate 2, leaving unnormalized expected residual error $1/6$; if it is one the world is known. Initial unnormalized risk is $5/6 + \varepsilon$. An optimal policy queries coordinate 2, then coordinate 3 if it is zero and coordinate 1 otherwise, identifying the world exactly. The ratio is $(4 + 6\varepsilon)/(5 + 6\varepsilon) \rightarrow 4/5 < 1 = B/(N - 1)$.

Theorem 6.3 (Sharp three-point, two-query constant). *[PROVED] For $N = 3$, $B = 2$, the worst-case ratio of sequential margin to the optimal adaptive policy is exactly $4/5$. This includes arbitrary random tie-breaking within the margin maximizers.*

Proof. We give a finite rational certificate proof; all certificate coefficients are displayed below, so the proof does not rely on floating-point optimization or a claimed exhaustive sample of priors. Use unnormalized risks. By coordinate flips suppose all marginal probabilities $p_i \leq 1/2$, and permute the first margin query to coordinate 3, so $p_3 \geq p_1, p_2$. Write the eight world probabilities in lexicographic order as $\pi = (\pi_{000}, \dots, \pi_{111})$.

For four masses in a 2 by 2 table, the smaller expected residual error after revealing one of the two coordinates is the sum of the two smallest masses. Indeed each row/column error selects two entries and is at least that sum; at least one of the two partitions separates the two largest entries, attaining it. Conditional on the first query, margin is optimal on the remaining two coordinates by Theorem 2.2. Thus its residual risk F_j after first querying j is the sum of the two smallest masses in each of the two j -slices.

For $j = 1, 2, 3$, form a 36 by 8 matrix $L^{(j)}$ as follows. List the indices of the four worlds with coordinate $j = 0$ and those with $j = 1$, each in lexicographic order. List their six two-element subsets in lexicographic order. For each pair of subsets, taking the zero-slice subset as the outer loop, put a row with ones on their union and zeros elsewhere. Then $F_j = \min_k L_k^{(j)} \pi$. The initial risk is $R_0 = \sum_i p_i$. It suffices to prove $5F_3 \leq R_0 + 4F_2$; symmetry gives the comparison with query 1 as well.

Let $z = (\pi, h) \geq 0$ and $e = (1, 1, 1, 1, 1, 1, 1, 0)$. Define the 41 by 9 matrix A by rows

$$A_k = (-L_k^{(3)}, 1) \ (1 \leq k \leq 36), \quad A_{36+i} = (x_i, 0) \ (1 \leq i \leq 3), \quad A_{40} = (x_1 - x_3, 0), \quad A_{41} = (x_2 - x_3, 0),$$

where x_i is the length-eight vector of coordinate- i labels in the world order. Let $b_{37} = b_{38} = b_{39} = 1/2$ and all other $b_i = 0$. At $h = F_3$ we have $Az \leq b$ and $e^\top z = 1$. For each $k = 1, \dots, 36$ set $c_k = (4L_k^{(2)} + x_1 + x_2 + x_3, -5)$. The table provides λ_k and a nonnegative vector v_k such that

$$c_k + A^\top v_k - \lambda_k e \geq 0, \quad d_k = \lambda_k - b^\top v_k \geq 0.$$

These are just nine rational inequalities and one rational equality per row, directly checkable using the matrix definitions. Therefore $c_k^\top z \geq \lambda_k e^\top z - v_k^\top Az \geq \lambda_k - v_k^\top b = d_k \geq 0$. Taking the minimum over k proves $5F_3 \leq R_0 + 4F_2$. Consequently $R_0 - F_3 \geq (4/5)(R_0 - \min_j F_j)$. The argument holds for every allowed first margin choice and its conditional continuation, hence also their randomized mixtures. Example 6.2 with $\varepsilon \downarrow 0$ proves sharpness. $\square$

k	λ_k	Nonzero entries of v_k (index:value)	d_k
1	1	1 : 3, 30 : 1, 35 : 1, 40 : 1	1
2	2	13 : 1, 15 : 1, 26 : 1, 27 : 1, 30 : 1, 37 : 1, 39 : 1	1
3	6/7	3 : 12/7, 5 : 8/7, 6 : 2/7, 27 : 12/7, 36 : 1/7, 41 : 5/7	6/7
4	4/5	13 : 7/5, 14 : 6/5, 17 : 3/5, 25 : 1/5, 35 : 8/5, 40 : 7/5	4/5
5	6/7	6 : 9/14, 9 : 1/2, 15 : 13/14, 17 : 15/14, 35 : 13/7, 40 : 31/14, 41 : 1/2	6/7
6	2/3	15 : 10/3, 29 : 1, 30 : 1/3, 35 : 1/3, 41 : 1	2/3
7	2	5 : 1, 11 : 1, 22 : 2, 35 : 1, 37 : 1, 39 : 1	1
8	5/2	1 : 3/4, 9 : 3/8, 10 : 3/8, 30 : 7/4, 32 : 3/8, 35 : 11/8, 39 : 3	1
9	3/2	5 : 3/2, 9 : 1, 24 : 2, 35 : 1/2, 39 : 3/2	3/4
10	2	13 : 1/3, 16 : 4/3, 17 : 1/3, 35 : 3, 38 : 1, 39 : 4/3	5/6
11	2/3	5 : 1/3, 10 : 4/3, 11 : 5/3, 35 : 5/3, 40 : 1/3	2/3
12	2	15 : 1/3, 18 : 5/3, 33 : 1/3, 35 : 8/3, 38 : 1, 39 : 5/3	2/3
13	2/3	4 : 10/3, 28 : 1, 35 : 2/3, 40 : 1/3	2/3
14	2	14 : 1/3, 16 : 4/3, 18 : 1/3, 30 : 3, 37 : 1, 39 : 4/3	5/6
15	2/3	4 : 4/3, 6 : 5/3, 12 : 1/3, 30 : 5/3, 41 : 1/3	2/3
16	1/2	16 : 2, 18 : 3/2, 28 : 1/2, 35 : 1, 40 : 1/2	1/2
17	0	4 : 1, 10 : 1, 18 : 2, 35 : 1, 40 : 1	0
18	1/2	18 : 7/2, 28 : 1, 35 : 1/2, 41 : 1/2	1/2
19	0	19 : 1, 20 : 2, 35 : 2	0
20	0	26 : 2, 27 : 1, 35 : 2	0
21	0	21 : 3, 35 : 2	0
22	0	19 : 1, 31 : 1, 35 : 3	0
23	0	20 : 3/2, 35 : 7/2, 40 : 1, 41 : 1/2	0
24	0	30 : 1, 32 : 1, 33 : 1, 35 : 2	0
25	0	19 : 1/2, 26 : 3/2, 35 : 3, 40 : 5/2	0
26	0	26 : 7/2, 35 : 3/2, 40 : 1/2	0
27	0	24 : 1, 26 : 3, 35 : 1, 40 : 1, 41 : 1	0
28	0	26 : 2, 35 : 3, 40 : 2, 41 : 1	0
29	0	26 : 3/2, 35 : 7/2, 40 : 5/2, 41 : 1/2	0
30	0	26 : 2, 30 : 3/2, 35 : 3/2, 40 : 1/2, 41 : 5/2	0
31	0	20 : 1, 22 : 1, 24 : 1, 35 : 2	0
32	0	26 : 1, 28 : 1, 30 : 2, 35 : 1	0
33	0	21 : 1, 24 : 2, 35 : 2	0
34	0	19 : 1, 34 : 1, 35 : 3	0
35	0	22 : 2, 35 : 3, 41 : 1	0
36	0	30 : 1, 32 : 1, 35 : 2, 36 : 1	0

Every displayed certificate was verified in exact rational arithmetic. The general finite- N, B sharp ratio is still unresolved. Random tie-breaking does not repair the unique-choice limiting construction.

Proposition 6.4 (What randomization does establish). *[PROVED] Uniform sampling without replacement achieves at least $(B/N)R_0$ expected reduction. With a known joint law, an optimal adaptive finite-horizon policy can be deterministic. In marginal-only one-step selection randomization can strictly improve the worst-case guarantee, as Theorem 3.1 proves.*

Proof. Each point is sampled with probability B/N and its original error is then zero; expected error at unqueried points cannot increase under Bayes conditioning. For known-law optimality, backward induction maximizes a linear function of the action probabilities at every history. $\square$

7 Misspecified selection and updating

For acquisition-regret comparisons, P is the reference posterior under the correct generating law at the fixed data state. A truth-knowing oracle conditions further on the realized world and is a different comparator. The risk identity also allows a point-mass P , but then its Bayes-optimal Hamming baseline is already zero.

Theorem 7.1 (True value under a model-based update). *[PROVED] Let P be the true joint law and Q the learner's law. For targets and observations as in Theorem 2.1, write $m_i^P, m_i^Q, c_{ij}^P, c_{ij}^Q$ for their moments. The learner initially predicts $h_i = \text{sgn}_+(m_i^Q)$ and, after $O_j = s$, predicts $g_{ij}(s) = \text{sgn}_+(m_i^Q + sc_{ij}^Q)$. Define $\alpha_{ij} = (g_{ij}(+1) + g_{ij}(-1))/2$, $\beta_{ij} = (g_{ij}(+1) - g_{ij}(-1))/2$. Its true post-query risk and gain from its own pre-query risk are*

$$F_{P,Q}(j) = \frac{1}{2} \sum_i w_i [1 - \alpha_{ij} m_i^P - \beta_{ij} c_{ij}^P], \quad (5)$$

$$U_{P,Q}(j) = \frac{1}{2} \sum_i w_i [(\alpha_{ij} - h_i) m_i^P + \beta_{ij} c_{ij}^P]. \quad (6)$$

These identities hold for any prescribed binary branch decisions g ; expressing g through Q moments assumes coherent conditioning under Q . On a Q -null observation branch the displayed sign convention specifies the learner's otherwise arbitrary action.

Proof. $g_{ij}(O_j) = \alpha_{ij} + \beta_{ij} O_j$ and $\mathbf{1}_{\{g \neq T_i\}} = (1 - T_i g)/2$. Take expectations under P and subtract the initial risk $\sum_i w_i (1 - h_i m_i^P)/2$. $\square$

Away from ties, inactive pairs $|c_{ij}^Q| \leq |m_i^Q|$ contribute zero gain. For active pairs,

$$U_{P,Q}(j) = \frac{1}{2} \sum_{i: |c_{ij}^Q| > |m_i^Q|} w_i [\text{sgn}_+(c_{ij}^Q) c_{ij}^P - \text{sgn}_+(m_i^Q) m_i^P].$$

Thus the sum on the right being zero exactly characterizes zero true value; it can also be negative. Positive P -optimal value is independently characterized by $|c_{ij}^P| > |m_i^P|$ for some i, j of positive weight.

Theorem 7.2 (Pairwise discrepancy bounds). *[PROVED] Define*

$$E_j = \frac{1}{2} \sum_i w_i (|m_i^P - m_i^Q| + |c_{ij}^P - c_{ij}^Q|), \quad D_j = \sum_i w_i \max\{|m_i^P - m_i^Q|, |c_{ij}^P - c_{ij}^Q|\}.$$

Let j_Q maximize Δ_Q and j_P maximize Δ_P . Then

$$\begin{aligned} |U_{P,Q}(j) - \Delta_Q(j)| &\leq E_j, & |\Delta_P(j) - \Delta_Q(j)| &\leq E_j, \\ \Delta_P(j_P) - U_{P,Q}(j_Q) &\leq E_{j_P} + E_{j_Q}, \\ 0 \leq F_{P,Q}(j_Q) - F_P(j_P) &\leq (D_{j_Q} + D_{j_P})/2 \leq \max_j D_j. \end{aligned}$$

The terminal-risk comparison uses a common endpoint and is preferable to comparing gains from two different baselines.

Proof. For the coherent sign rule, either the branch is constant and equals h_i (zero gain), or $\alpha_{ij} = 0$, $|\beta_{ij}| = 1$, so $|\alpha_{ij} - h_i| \leq 1$. Apply (6); for Δ apply the Lipschitz argument in Corollary 2.3. Maximization gives the gain comparison. For terminal risks exactly one of α, β has magnitude one, yielding $|F_{P,Q}(j) - F_Q(j)| \leq D_j/2$. Also $|F_P(j) - F_Q(j)| \leq D_j/2$ because a maximum is 1-Lipschitz in the sup norm. Insert $F_Q(j_Q) \leq F_Q(j_P)$. Nonnegativity follows from Bayes optimality under P . $\square$

Corollary 7.3 (Approximation and refit defect). **[PROVED]** *If the estimated Q scores have uniform error at most ϵ , the terminal bound acquires 2ϵ . If the actual branch predictor $\tilde{g}$ differs from the coherent predictor with weighted probability*

$$\zeta_j = \sum_i w_i P_P(\tilde{g}_{ij}(O_j) \neq g_{ij}(O_j)),$$

add ζ_j for the selected candidate.

Proof. Score maximization loses at most 2ϵ . Two predictions' zero-one losses differ by at most their disagreement indicator. Sum and take expectations. $\square$

Counterexample 7.4 (Selection failure with identical marginals). **[COUNTEREXAMPLE]** Let all variables be fair. Under P , O_1 is independent of T and $O_2 = T$; under Q , swap the two observations. The unique Q -optimal query has zero true value, while the P -optimal query has value $1/2$. All first moments agree exactly.

Counterexample 7.5 (Zero value even in the full pool). **[COUNTEREXAMPLE]** Take a fair root R and two leaves. Under P let the leaves be $-RV_1, -RV_2$, with probabilities $1/2, 1/4, 1/4$ on $(V_1, V_2) = (++, (+-), (-+))$. Under Q let them be RV_1, RV_2 with probabilities $3/5, 1/5, 1/5$. All three marginals are fair. Under P root-leaf moments are $-1/2$ and the leaf-pair moment is zero. Under Q they are $3/5, 3/5, 1/5$. The unique Q -optimal query is the root (column sum $11/5$ versus $9/5$). Its Q update predicts all labels equal to the root. True posterior risk is $(0 + 3/4 + 3/4)/3 = 1/2$, exactly the baseline: zero gain. The P -optimal query is also the root, but its correct update gains $1/3$. This isolates update failure from candidate misranking.

[NUMERICALLY CHECKED] Finite random-law checks for $N = 2, \dots, 6$ (1,500 P, Q pairs) confirmed the direct-risk identities, terminal discrepancy bound, and common-noise margin inequality. These checks supplement the proofs. **[NUMERICALLY CHECKED]** A separate fixed-marginal check for $(.1, .1, .25)$ enumerated the six vertices of its world-law polytope and 50,000 random convex mixtures. The smallest tested ratio of the selector in Theorem 4.4 was $20/23$, attained at a vertex. The proof, rather than this finite sample, establishes the universal claim.

8 Week 15 cross-audit and corrected propositions

Inspected: `outputs/week15_boundary_acquisition/WEEK15_REPORT.md`, `THEORY_WEEK15.md`, `THEORY_ERRATA.md`, the Week 14 theory, and `src/week15_bayes_frontier.py`, `week15_metric.py`, `week15_metric_confirm.py`. Empirical results were not rerun.

Item	Verdict	Minimal correction
B1	Correct identity; gap	XOR and orientation invariance are exact. Error-set perimeter is not a displacement metric; delete its general identification with NSD/ASSD.
B2	Gap	Fixed- r , fixed-set iid-cloud consistency is correct. Fix the posterior independently of the evaluation cloud or prove uniform convergence. Require positive denominator. Fixed- k kNN and shrinking-radius claims need separate normalization/assumptions. Graph Dice is not NSD.

Item	Verdict	Minimal correction
B3	Wrong as exact; re-pairable	An unspecified spread does not imply small overlap probability. State a posterior law. For a uniform threshold posterior, normalized plug-in loss is $1 - r/(3s)$, not exactly one. Static risk does not prove the cause of an acquisition run's collapse.
B4	Correct theorem; empirical gap	Fix state, posterior, metric, update, and coupling of potential labels. Truth-information gap is $V_E - V_B$; $V_E - V_{\text{margin}}$ also contains a method gap. Separate-policy AULCs are not this decomposition. The inspected implementation is Laplace/Monte Carlo NSD look-ahead.
M1	Correct core; gaps	For true weights and fixed radius/geometry, bounded-below measurable density suffices. Smoothness at graph scale is unnecessary. Assume uniform density consistency explicitly; the listed k conditions do not establish it. Fixed 95% clipping changes the estimand. Radius and standardization also vary between designs in the implementation.
E15-1	Correct	Exchangeable fixed counts, distinct queries, independent randomization, both classes present; survival at zero is one.
E15-2	Correct number; notation gap	Use $(H_{108}^2 + \sum_{i=1}^{108} i^{-2})/2 = 14.67254557\dots$. Standard $H_n^{(2)}$ alone usually means $\sum i^{-2}$.
E15-3	Correct with convention	Resolve ties and compute AUC on the same order. Half-credit AUC on unresolved ties can violate the claimed relation to the query order.
E15-4	Correct	The convex dominance closure is contained in the implied set. State consistency and strict-inequality conventions for the two feasibility LPs.
E15-5	Correct	The zero-budget survival term is one; the binomial sum starts at one.

Proposition 8.1 (Fixed-cloud continuum estimands). *[PROVED] Let X_i be iid on bounded Ω with density $p \geq p_{\min} > 0$, and let h be a bounded measurable symmetric pair kernel, fixed independently of the cloud. Then, almost surely,*

$$\frac{1}{\binom{n}{2}} \sum_{i < j} \frac{h(X_i, X_j)}{p(X_i)p(X_j)} \rightarrow \int_{\Omega} \int_{\Omega} h(x, y) dx dy.$$

This applies to fixed-radius true, predicted, shared and mismatched cuts. Ratios converge when their limiting denominators are positive. A fixed posterior average independent of the cloud is allowed. If $\|\hat{p} - p\|_{\infty} \rightarrow 0$ almost surely, plug-in weights have the same limit.

Proof. The bounded weighted kernel obeys the U-statistic strong law; sampling densities cancel. For posterior averaging use its integrated kernel, conditioning first on fixed training data. Uniform density consistency implies eventual $\hat{p} \geq p_{\min}/2$ and uniformly converging inverse products. Use continuous mapping for the ratio. $\square$

The unweighted uniform-cloud limit has factor $|\Omega|^{-2}$. Density continuity is unnecessary for true weights. If clipping thresholds converge to a finite c , the limit is instead $\int \int h(x, y) \min\{1, c p(x)p(y)\} dx dy$ by dominated convergence. Fixed quantile clipping therefore generally retains density dependence.

Proposition 8.2 (Exact flat-boundary repair). *[PROVED] On $(-L, L)$ let $A_{\theta} = \{x > \theta\}$, with all thresholds farther than r from the endpoints. For ordered pairs with $|x - y| < r$,*

$$J_r(A_{\theta}) = r^2, \quad S_r(A_{\theta}, A_0) = (r - |\theta|)_+^2.$$

If $\Theta \sim \text{Uniform}[-s, s]$, $s \geq r$, $L > s + r$, then

$$\mathbb{E}S_r = r^3/(3s), \quad \mathbb{E}J_r(A_{\Theta} \triangle A_0) = 2r^2(1 - r/(3s)).$$

A constant predictor has mismatch r^2 , strictly smaller. Normalized ratio-of-expectations losses are $1 - r/(3s)$ and one, respectively.

Proof. Crossing pairs form two triangles of total area r^2 ; shared crossings form triangles with side $(r - |\theta|)_+$. Integrate the square against the uniform density and use mismatch = true + predicted - 2 shared. $\square$

[COUNTEREXAMPLE] For $\theta = r/2$, graph Dice is $1/4$ while NSD with tolerance r is one. These functionals differ. **[PROVED]** If expected true cut mass is positive, normalized static loss is at most one for every predictor because shared mass is nonnegative. A constant predictor equals one, so it is never strictly preferred over a predictor with positive shared mass. This does not prove NSD improvement by the resulting acquisition.

Proposition 8.3 (Oracle decomposition). **[PROVED]** Fix data, a finite common action set, a joint law of truth f and all potential labels (Y_c) , and an integrable gain $G(c, Y_c, f)$ for a fixed update procedure. Then

$$V_R = \mathbb{E} \max_c G(c, Y_c, f) \geq V_E = \mathbb{E}_f \max_c \mathbb{E}[G(c, Y_c, f) \mid f] \geq V_B = \max_c \mathbb{E} G(c, Y_c, f).$$

V_B dominates any rule independent of unknown truth and potential labels conditional on the data.

Proof. Apply convexity of the maximum to conditional expectation twice. A non-anticipating randomized rule averages fixed action values. $\square$

Label marginals alone do not define V_R ; a coupling is required. The chain holds for any such coupling. Values need not be nonnegative for a fixed non-Bayes prediction/update procedure evaluated by NSD.

[NUMERICALLY CHECKED] The inspected frontier code uses $\mathbf{S}=24$, Laplace Gaussian posterior samples, and fresh fantasy-label Laplace fits. A well-specified prior does not make these exact posterior updates. The reported $0.0155/0.0645 \simeq 24\%$ compares absolute gains. Relative to reported margin gain -0.007 , captured headroom is $(0.0155 + 0.007)/(0.0645 + 0.007) \simeq 31.5\%$. Both descriptions must name their denominator.

Proposition 8.4 (Precise errata statements). **[PROVED]** For n iid points with independent continuous coordinates in three dimensions, expected minimal Pareto-front size is $\sum_{i=1}^n H_i/i = (H_n^2 + \sum_{i=1}^n i^{-2})/2$. In a strict score order with r rare and m common labels, let a_i count common labels preceding rare label i , in rare-label order. First rare position is $a_1 + 1$ and $m(1 - \text{AUC}) = r^{-1} \sum_i a_i \geq a_1$. Equality holds iff rare labels are consecutive.

Proof. Sort in the first coordinate. Point i is minimal iff its two other coordinates are undominated among the first i points. Their expected minimal count is H_i , by record probabilities, so exchangeability gives probability H_i/i . Sum and rearrange the harmonic double sum. For AUC each rare–common inversion contributes one pair. The nondecreasing a_i have average equal to their minimum iff all equal. $\square$

[COUNTEREXAMPLE] Positive point $(1, 0)$ and negative point $(0, 1)$ under positive linear weights imply $w_1 > w_2 > 0$, $t \in [w_2, w_1]$. Point $(2, -1/2)$ must be positive but is outside the upward closure of the positive convex hull, proving E15-4’s containment can be strict. **[PROVED]** Exact implication is two candidate-label feasibility checks. For finitely many constraints, strictly positive weights and positive-class margins can be jointly scaled to at least one; negative scores remain at most zero. These are LP constraints. Exclude inconsistent training data.

9 Round-1 results retained: prediction, discovery, and stopping

Counterexample 9.1 (Better marginals, worse acquisition). **[COUNTEREXAMPLE]** Use a fair decoy and $M = N - 1$ copies of Bernoulli($1/2 + \eta$), $N \geq 3$. Model A reports true probabilities. Model B reports $1/2 + 2\eta$ on the decoy, $1/2 - \eta/2$ on one cluster point and $1/2 + 3\eta/2$ on the others,

for small $\eta > 0$. A is strictly better in expected Brier/log loss at every point and has classification error lower by $2\eta/N$. B's average Brier excess is $\eta^2(M+24)/(4(M+1))$ and maximum probability error 2η . Yet A's margin queries the decoy and B's queries the cluster. Under a common ideal update their acquisition-value ratio tends to $1/M$.

Proof. Proper scores minimize uniquely at the true probabilities. B changes the majority decision on one cluster point only. Its squared errors sum to $4\eta^2 + (3\eta/2)^2 + (M-1)(\eta/2)^2$. Its smallest absolute margin is the misclassified cluster point. $\square$

Theorem 9.2 (Finite-population discovery). *[PROVED] In a uniform random order of $n_0, n_1 \geq 1$ labels, $N = n_0 + n_1$, let T be first both-class discovery. Then*

$$\mathbb{P}(T > 0) = 1, \quad \mathbb{P}(T > k) = \frac{\binom{n_0}{k} + \binom{n_1}{k}}{\binom{N}{k}} \quad (k \geq 1), \quad \mathbb{E}T = \frac{N+1}{n_0+1} + \frac{N+1}{n_1+1} - 1.$$

For first discovery D of a designated class with r members,

$$\mathbb{P}(D > k) = \frac{\binom{N-r}{k}}{\binom{N}{k}} = \frac{\binom{N-k}{r}}{\binom{N}{r}} \leq \min\{(1-r/N)^k, (1-k/N)^r\}, \quad \mathbb{E}D = \frac{N+1}{r+1}.$$

The geometric mean N/r exceeds this by $(N-r)/(r(r+1))$.

Proof. Survival means a monochromatic sample; its two events are disjoint for $k \geq 1$. If D_0, D_1 are first class positions, $\min(D_0, D_1) = 1$, hence $T = D_0 + D_1 - 1$. The minimum of r uniform distinct positions has expectation $(N+1)/(r+1)$, by exchangeability of the $r+1$ gaps. The two survival product forms give the bounds. $\square$

Corollary 9.3 (Exchangeable minimax and limits). *[PROVED] Under the uniform fixed-count assignment prior, all adaptive distinct query rules with truth-independent randomization have this discovery law. Uniform query ordering is minimax over assignments for expected discovery time. For fixed r , $T/N \Rightarrow \text{Beta}(1, r)$. If $r \rightarrow \infty$, $r/N \rightarrow 0$, then $(r/N)T \Rightarrow \text{Exp}(1)$.*

Proof. Unqueried labels remain exchangeable after every history; the transition probabilities depend only on remaining counts. A policy's worst assignment is at least its prior average, and uniform ordering equalizes assignments. The rare-class survival products tend to $(1-x)^r$ and e^{-x} at the respective scalings. Except with probability r/N , the first label is common and T equals rare-class discovery time. $\square$

[NUMERICALLY CHECKED] At $N = 136, r = 12$: $\mathbb{E}T = 10.6344615\dots$, $P(D > 9) = 0.42394419\dots$, $P(D > 16) = 0.2078709166\dots$; rare-class 95% and 99% discovery quantiles are 29 and 42. These are design calculations, not a frozen thesis endpoint estimate.

Proposition 9.4 (Exact enrichment condition). *[PROVED] A subset of size M containing $s \geq 1$ designated-class members has first-discovery time stochastically no larger than a full pool of size N containing r , iff $s/M \geq r/N$. Both-class startup additionally needs an opposite-class anchor or a separate argument.*

Proof. Necessity is the first-draw probability. For sufficiency compare hazards $s/(M-j)$ and $r/(N-j)$ while survival is possible: $sN - rM + j(r-s) \geq 0$, because the subset has $s \leq r$. Multiply the complementary hazards; subset exhaustion sets survival to zero. $\square$

Counterexample 9.5 (Weak global prediction can aid startup). **[COUNTEREXAMPLE]** Let $N = r^2$, $r \geq 4$. A three-point high-score subset contains one rare and two common labels; outside it are $r-1$ rare labels. Both conditional rare probabilities are below $1/2$, so the Bayes classifier remains constant. Rare-oriented AUC is $1/2 + \frac{1}{2}(1/r - 2/(N-r)) \rightarrow 1/2$. Calibrated Brier improvement is $(N-3r)^2/[3N^2(N-3)] \rightarrow 0$ and log-loss gain is at most $h_2(3/N) \rightarrow 0$. Both-class discovery within the subset has mean $7/3$; full-pool mean is asymptotic to r .

Proof. Count binary-score pairs for AUC; compute the variance of the conditional rare-class probability for Brier improvement. Mutual information is bounded by the subset indicator entropy. Apply the discovery formula to each pool. $\square$

Counterexample 9.6 (Same startup time, different information). **[COUNTEREXAMPLE]** For fair Θ , let labels be $(0, 1, \Theta, 1 - \Theta)$. Querying the first two or last two gives $T = 2$ in both cases, but posterior full-pool risk is respectively $1/4$ or zero.

Theorem 9.7 (Stopping-time information). **[PROVED]** Let Θ be a finite identifiable family of deterministic binary worlds, each containing both classes. Let U be an independent policy seed, H_T the history at first both-class discovery, and $Q \geq T$ a zero-error identification stopping time. In bits,

$$I(\Theta; H_T \mid U) = H(Y_{\text{first}}, T \mid U) \leq 1 + H(T), \quad \mathbb{E}Q \geq \mathbb{E}T + H(\Theta) - 1 - H(T).$$

If $T \leq b$, the information is at most $\log_2[2(b-1)]$; if T is deterministic it is at most one bit.

Proof. Given U , all labels before T equal the first label and the last label is opposite. Thus (Y_{first}, T) reconstructs all queries and labels; conversely the history gives this pair. History is deterministic given (Θ, U) , proving the information identity. Conditional on (H_T, U) the remaining zero-error decision tree is a binary prefix code, with mean length at least $H(\Theta \mid H_T, U)$. Average and use independence of U . There are at most $2(b-1)$ pairs. $\square$

10 Round-1 results retained: boundary metrics and structural priors

Counterexample 10.1 (A nearest-opposite-label band can reverse boundary ranking). **[COUNTEREXAMPLE]** Let $m \geq 4$, $0 < \epsilon < 1/(5m)$, and use the $5m$ test points $-2, -\epsilon, \epsilon, 2\epsilon, \dots, (5m-2)\epsilon$ with truth $y(x) = \mathbf{1}_{\{x>0\}}$. Let q_{20} be the m points of smallest distance to an opposite-labelled test point. Then $q_{20} = \{-\epsilon, \epsilon, \dots, (m-1)\epsilon\}$, with a strict cutoff. Threshold A at -1 has band accuracy $(m-1)/m$ and boundary error one. Threshold B at $(m-3/2)\epsilon$ has band accuracy $2/m$ and boundary error $(m-3/2)\epsilon$. Both have nonempty boundaries; the band has both classes. Taking $\epsilon = 1/(100m^2)$ makes the geometric reversal arbitrarily large. Embed the construction in four dimensions by constant other coordinates; a common affine scaling of the varying coordinate preserves it.

Proof. Nearest-opposite distances are $2+\epsilon$ at -2 , 2ϵ at $-\epsilon$, and $(j+1)\epsilon$ at $j\epsilon$. They give the stated strict band cutoff. A misclassifies only its negative band point. B correctly classifies that point and the largest positive band point. Distances of the thresholds from zero give the geometric errors. Training anchors at $\pm R$, $R > 4$, can be included without changing this test-band construction. $\square$

Proposition 10.2 (Graph cuts and localization). **[PROVED]** On a graph $G = (V, E)$ let $C_y = \{\{i, j\} \in E : y_i \neq y_j\}$ and $A = \{i : y_i \neq \hat{y}_i\}$. Then $C_y \Delta C_{\hat{y}} = \delta_G(A)$. Zero cut mismatch means the two labelings differ by an independent global flip on each connected component; one known label per component removes this ambiguity. Also $|C_y \Delta C_{\hat{y}}| \leq d_{\max}|A|$. Cut disagreement count alone cannot measure displacement.

Proof. For each edge, $(y_i \oplus y_j) \oplus (\hat{y}_i \oplus \hat{y}_j) = (y_i \oplus \hat{y}_i) \oplus (y_j \oplus \hat{y}_j)$. An error set with no crossing edge is a union of connected components. Counting incident edges gives the degree bound. On a path, any two different single-cut labelings disagree on exactly two cut edges, irrespective of their separation. $\square$

Proposition 10.3 (A finite-pool boundary-location metric). **[PROVED]** For geometric edges define

$$\rho(\{x_1, x_2\}, \{y_1, y_2\}) = \min_{\pi \in \mathfrak{S}_2} \max_{i=1,2} \|x_i - y_{\pi(i)}\|.$$

Hausdorff distance induced by ρ is a metric on nonempty finite edge sets. Add the empty set by assigning its distance to every nonempty set a constant $D_0 \geq \text{diam}(\Omega) > 0$. Applied to cut sets it is a metric on labelings modulo component flips; anchors give identity of labelings. On an equally spaced path, two single cuts i, j have distance $|i - j|h$.

Proof. The optimal endpoint matchings compose, proving the triangle inequality for ρ ; the other metric properties follow from endpoint matching. Hausdorff distance inherits these properties on finite sets. All nonempty distances are at most $\text{diam}(\Omega)$, so adding the empty set preserves the triangle inequality. Use Proposition 10.2 for the quotient. For path edges, matching left to left and right to right minimizes the maximal displacement. $\square$

Theorem 10.4 (Geometric approximation by graph cuts). *[PROVED] Let Ω be bounded and convex and X an ϵ -net. Connect all pairs at distance at most 6ϵ . Let A, B have compact smooth boundaries with two-sided tubular neighborhoods of radius strictly greater than 2ϵ , contained in Ω . Label X by membership in each set. For their cut sets,*

$$|H_\rho(C_A, C_B) - H(\partial A, \partial B)| \leq 12\epsilon.$$

Proof. At each boundary point take normal offsets $\pm 2\epsilon$ and net points within ϵ of them. The tubular condition puts these net points on opposite sides; their distance is at most 6ϵ and their edge midpoint is within 3ϵ of the boundary point. Conversely every crossing edge meets the boundary along its segment (convexity keeps the segment in Ω), so its midpoint is within 3ϵ of a boundary point. Each midpoint set is therefore within Hausdorff distance 3ϵ of its boundary. Their mutual Hausdorff distance differs from the true one by at most 6ϵ . For two edges of length at most 6ϵ , midpoint distance is at most ρ , which is at most midpoint distance plus 6ϵ . The same comparison holds for Hausdorff distances, giving 12ϵ . $\square$

Proposition 10.5 (Moment hierarchy). *[PROVED] Current Bayes vertex Hamming risk uses first moments; one-query vertex Hamming value uses second moments. Current cut-Hamming risk for the edgewise Bayes cut predictor uses second label moments. One-query cut-Hamming value can require third moments.*

Proof. The first two claims follow from Theorem 2.1. The spin encoding a cut on $\{i, j\}$ is $-S_i S_j$, so its mean is a second moment and its cross-moment with a queried S_k is a third moment. For strict necessity compare three independent fair bits with the uniform law on 000, 011, 101, 110. All first and second moments agree. On the triangle the current edgewise Bayes cut risk is $3/2$ for both. After one bit, independent bits give zero reduction, whereas even parity determines the opposite edge and gives reduction $1/2$. The edgewise predictor need not itself be realizable as a vertex cut; this distinction is part of the stated loss/action model. $\square$

Proposition 10.6 (A physics prior can help one target distribution and hurt another). *[PROVED] At two independent design sites minimize $\sum_i n_i (g_i - f_i)^2 + \lambda \sum_i (g_i - \mu_i)^2$. Then $g_i - f_i = \lambda(\mu_i - f_i)/(n_i + \lambda)$. Compared with the zero prior, weighted squared-risk change is*

$$\sum_i q_i \left(\frac{\lambda}{n_i + \lambda} \right)^2 (\mu_i^2 - 2\mu_i f_i).$$

For $f = (10, 1)$, $\mu = (10, 3)$, $n = (9, 1)$, $\lambda = 1$, target weights $(.9, .1)$ give risks $.4, .1, .925$ for the physics prior alone, physics plus correction, and zero-prior correction. Weights $(.1, .9)$ give $3.6, .9, .325$ respectively.

Proof. Differentiate the strictly convex objective coordinatewise, substitute the solution, and expand the squared-error difference. The stated risks follow by direct substitution. $\square$

Proposition 10.7 (Smoothness alone does not exclude finite sign worlds). *[PROVED] On distinct finite locations with minimum spacing $d_{\min}$, every binary sign assignment is realized by an L -Lipschitz function, for every $L > 0$. It is also realized by a function of arbitrarily small positive norm bound in an RKHS whose pool Gram matrix is strictly positive definite.*

Proof. Assign values αs_i with $0 < 2\alpha \leq Ld_{\min}$ and extend by the McShane formula $\min_i(\alpha s_i + L\|x - x_i\|)$. For an RKHS with Gram matrix K , the interpolant with coefficient vector $K^{-1}\alpha s$ has squared norm $\alpha^2 s^\top K^{-1}s$, which can be made arbitrarily small. $\square$

Known amplitude exceeding L times a coverage radius can certify a sign; binary signs alone do not supply that amplitude. These are elementary extension/interpolation facts, not novelty claims.

11 Concrete predictions for a Laplace GP classifier

Let $Q_f = \mathcal{N}(\mu, \Sigma)$ be the current Laplace latent approximation. Two target definitions must be separated. For *persistent observed labels*, draw conditionally independent Y_i once given f , with $P(Y_i = 1 \mid f) = \sigma(f_i)$, and reveal that same Y_j on query. Then for distinct indices,

$$m_i = \mathbb{E}_{Q_f} \tanh(f_i/2), \quad c_{ij} = \mathbb{E}_{Q_f} \tanh(f_i/2) \tanh(f_j/2), \quad c_{jj} = 1.$$

For *latent-sign targets* $T_i = \text{sign}(f_i)$ and a logistic observation O_j , instead use

$$m_i = \mathbb{E}_{Q_f} \text{sign}(f_i), \quad c_{ij} = \mathbb{E}_{Q_f} \text{sign}(f_i) \tanh(f_j/2).$$

In particular $c_{jj} = \mathbb{E}|\tanh(f_j/2)| < 1$ for finite nonsingular latents. These moments are computable by one- or two-dimensional Gaussian quadrature. Coherent fantasy conditioning reweights Q_f by the observation likelihood; a fresh Laplace fit generally gives a different law.

Each row below gives a falsifiable calculation, not an assertion that a GP can realize every extremal binary law or that Hamming rankings predict NSD.

Result	Computable prediction / next check
Binary identity, Theorem 2.1	For every candidate, quadrature moments and direct coherent two-branch posterior Hamming risks must agree. Test persistent-label and latent-sign targets separately; substituting latent covariance for c_{ij} should not be expected to agree.
Sharp margin and balanced κ_N	For persistent-label targets and coherent updating, verify $\Delta_{\text{margin}}/\Delta^* \geq 1/(N-1)$. At zero latent means, all marginals are fair; verify uniform ratio $\geq \kappa_N$. Failure beyond numerical tolerance diagnoses inconsistent moments, targets, or updates. Attainment by a particular GP kernel is not predicted.
General/two-level marginals, Theorem 4.2	On a small pool, compute moments and the finite LP using only the marginal vector. Its worst-law guarantee must hold for the GP joint law. At two points with unequal uncertainties, margin must be Hamming-optimal under coherent revelation.
Exact two-level family, Theorem 4.4	Tune a three-point Gaussian latent law's marginal means to persistent-label probabilities (.1, .1, .25). Query with probabilities (5/46, 5/46, 18/23); the expected coherent Hamming gain divided by the best candidate gain must be at least 20/23. A GP-restricted optimum may be better than this universal-law guarantee.
Rare marginals, Theorem 4.5	Use a constant-mean equal-variance latent prior tuned so every label marginal equals 3/34. For $N = 136$, its uniform-query ratio cannot fall below 2798/86615. Compare with the constant-count posterior separately. The value is a universal-law lower bound, not a claim that this GP attains it.

Result	Computable prediction / next check
Positive transfer, Proposition 5.1	For latent-sign targets and logistic observations, mark candidates with no pair satisfying $ c_{ij} > m_i $. Their exact one-step Bayes Hamming gain must be zero despite positive mutual information or observation entropy.
Common/heterogeneous noise, Theorem 5.3	Add a common independent BSC channel to persistent labels and verify the margin bound using ηc_{ij} . With candidate-dependent noise, search for margin candidates with no decision-crossing pair; positive information alone will not prevent zero gain.
Budgets, Theorem 6.1	On a small finite label law induced by the GP, condition exactly and compare margin with dynamic programming for $B = 2, 3$. Verify the B/N bound. For $N = 3, B = 2$, verify the stronger sharp $4/5$ bound from Theorem 6.3. Refitting a Laplace model is a separate experiment and need not obey these bounds.
Misspecification, Theorems 7.1–7.2	In a synthetic GP world with a controlled reference posterior P , compare direct branch risks with the P, Q moment formula; terminal regret must be at most $\max_j D_j$. Identical marginal calibration does not remove the $c^P - c^Q$ term. A single realized ground-truth function is not itself this posterior P .
Update defect	At fixed states, compare coherent likelihood reweighting of the current Laplace Gaussian with fresh Laplace fantasy fits. Record weighted decision disagreement ζ_j and verify its risk bound. Repeat numerical integration/Monte Carlo precision checks before ranking the 24-sample frontier estimates.
B1–B3, graph versus displacement	For GP posterior draws near a flat interface, vary posterior location spread and graph radius separately. The uniform-threshold control must recover $1 - r/(3s)$ normalized loss. Plot graph Dice and NSD on the same displaced thresholds; the analytic discrepancy at displacement $r/2$ is $1/4$ versus one.
B4, oracle decomposition	At the same frozen states, use a common coupled fantasy-label array and the same refit/metric rule for all oracle levels. Population estimates should satisfy the Jensen chain as sampling error shrinks; separate policy AULCs need not form an exact decomposition.
M1, Proposition 8.1	Freeze scaling and radius; with known densities and no clipping, replicate cloud sampling and check convergence to one common pair-integral Dice. Then change clipping, density estimation, and radius one at a time. Fixed clipping is predicted to retain density dependence.
Errata / discovery	Under exchangeable fixed-count posterior worlds, feature-adaptive queries cannot improve discovery over random. Verify $t = 0$ separately and reproduce the 14.6725 front calculation as an arithmetic control, not as evidence for a GP acquisition improvement.

12 What remains unresolved

[CONJECTURE] The exact general finite- N, B sequential-margin competitive ratio remains open here. The $N = 3, B = 2$ case is now proved to be $4/5$ by the rational certificates in Theorem 6.3. The general-marginal LP is exact but may be computationally prohibitive; the compact rare-marginal formula currently covers homogeneous $p \leq 1/3$ (and $p \geq 2/3$ by flipping labels). There is no claim of a sharp GP-restricted minimax constant, a universal NSD margin ratio, or a publication-priority theorem. The user’s independent round-1 checks and the scratch numerical checks support specified formulas only; they do not resolve these open questions. The requested Astra-ultra subagents encountered their usage limit; the extensions assembled here did not receive a completed independent second proof audit. Source environments and references passed static checks. The native LaTeX compiler returned a platform-directory error (**Unable to find standard directories for platform**); PDF compilation and visual layout therefore remain unverified.

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